

WEARABLE ENERGY HARVESTER FOR HEALTH ANALYSIS FROM FOOT MOTION

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

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BONAFIDE RECORD OF WORK DONE BY

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SYNOPSIS

Most wearable health and fitness gadgets need a constant power supply to function — and the battery problem is one that everyone using these devices runs into sooner or later. Frequent recharging is inconvenient, and in long-term monitoring scenarios, it can seriously limit how useful these devices actually are. Walking, on the other hand, is something people do all the time, and it turns out there is quite a bit of mechanical energy going to waste with every step. This project explores whether that wasted energy can be put to use by building a shoe-integrated piezoelectric harvesting system.

The proposed system utilizes a piezoelectric sensor (At the core of the design is an LDT2-028K piezoelectric sensor fixed onto a cantilever beam that sits on the outside of the shoe. Each time the foot lands, the vibrations make the cantilever flex, and that flexing is what produces the electrical output — an alternating voltage, typically somewhere between 1–1.2 V (open circuit), with current in the microampere range, indicating low-power energy generation suitable for wearable applications.

Since the raw output from the piezoelectric element is AC, it needs to go through a DB107 bridge rectifier before it can be stored or used. The rectified DC is then held in a capacitor. Managing how this stored energy gets used is where the MOSFET-based switching circuit comes in — it keeps an eye on the capacitor voltage and connects the load to the harvested supply when the voltage climbs past 2.4 V. If the voltage falls below that, it quietly switches over to a rechargeable battery, so nothing drops out.

A printed circuit board (PCB) has been designed to integrate the rectification and switching circuits into a compact module. The current prototype demonstrates the feasibility of harvesting energy from low-frequency human motion and managing it effectively. The integration of loads such as a step counter and motion sensor (MPU) is planned as future work.

This project presents a simple, low-cost, and practical approach to piezoelectric energy harvesting, contributing towards the development of self-powered wearable systems and sustainable energy solutions.

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CHAPTER 1

INTRODUCTION

Wearable devices for health monitoring have quietly become part of everyday life over the last decade. Whether it is a fitness band counting steps or a sensor tracking heart rate through the night, these gadgets are useful — but they all share the same weakness: they run on batteries. Replacing or recharging them interrupts continuous monitoring and adds friction for the user. What does not get talked about as much is how much energy the human body produces just by going about its day. Walking alone generates a surprisingly large amount of mechanical energy that simply gets absorbed by the ground and lost. Finding a way to recover even a fraction of that energy could make a real difference for low-power wearable devices.

Harvesting energy from body motion is not a new idea, but it has gotten much more practical as piezoelectric materials and low-power electronics have improved. Piezoelectric harvesting works by taking mechanical deformation — bending, pressing, vibrating — and turning it directly into electrical charge. That makes it a natural fit for something like footstep-induced vibrations, which are low in frequency but reasonably consistent as long as someone is walking.

For this project, we built a cantilever-based harvesting assembly that mounts on the outside of a shoe. The LDT2-028K piezoelectric sensor is bonded to the cantilever strip, and every footfall makes it flex back and forth, generating an AC voltage. That signal is not directly usable, so it gets passed through a bridge rectifier to produce DC and then stored in a capacitor until there is enough charge to be worth doing something with.

Getting the energy out of the capacitor and to the load in a controlled way required a dedicated power management stage. We used a MOSFET-based switching circuit that checks the capacitor voltage and decides whether to feed the load from harvested energy or fall back on a rechargeable battery. The handover between the two sources is nearly seamless, which means the load keeps running regardless of whether someone is actively walking.

What we ended up with is a fairly simple and inexpensive system that actually works. It may not produce enough power to run a smartphone, but for small sensors and counters, it holds up. Combining the harvesting and management sides of the problem in one compact board is, we think, a useful step toward wearables that can genuinely sustain themselves from everyday movement.

1.1 PROBLEM STATEMENT

Wearable health monitoring devices have moved from novelty to necessity in a surprisingly short time. The problem is that nearly all of them run on batteries, and batteries need attention — charging, replacing, managing. For a device that is supposed to run quietly in the background for days or weeks at a time, that is a meaningful limitation. In clinical or home-care settings where uninterrupted monitoring matters, even a short gap caused by a dead battery can be consequential.

Meanwhile, every person walking around is producing mechanical energy with each step — energy that currently does nothing but compress the ground. There have been attempts to harvest this, but many existing solutions are either too bulky to wear comfortably, too complex to be worth the effort, or simply not efficient enough at the slow, irregular frequencies that characterize normal walking.

Even when energy can be successfully extracted from foot motion, the output is inherently small and arrives in bursts rather than as a steady stream. Without some form of power management sitting between the harvester and the load, that energy would be nearly impossible to use reliably. This is a part of the problem that often gets overlooked in simpler harvesting designs.

What is actually needed is a system that addresses all three parts of the problem together: capturing energy from walking, storing it properly, and deciding when and how to use it. It also has to be small enough and cheap enough to be worth building into a wearable product. That is the gap this project aims to fill.

1.2 OBJECTIVE

- Design and build a cantilever-based piezoelectric harvesting assembly that can capture the vibrations produced during normal human walking.
- Use the LDT2-028K piezoelectric sensor to convert the mechanical strain from cantilever bending into an electrical signal.
- Design a rectification stage using the DB107 bridge rectifier to turn the alternating sensor output into usable DC.
- Build a capacitor-based storage stage that accumulates the rectified energy across multiple walking cycles.
- Implement a MOSFET switching circuit that manages the transition between harvested energy and battery backup based on the stored voltage level.
- Design and fabricate a compact PCB in KiCad that integrates the rectifier and switching circuitry into a single board.
- Show that a practical, low-cost harvesting system for slow human motion frequencies is achievable with off-the-shelf components.
- Lay the groundwork for connecting real wearable loads — step counters, MPU sensors — to the harvested power supply in future iterations.

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORK

A fair amount of research has gone into the question of how to power wearable devices from the body itself rather than from a conventional battery. Walking turns out to be one of the more promising sources, and the approaches that have been tried including piezoelectric, triboelectric, and hybrid systems — each with its own set of trade-offs.

Iftikhar Ahmad and colleagues (2025) put together a hybrid harvester built into footwear, combining a PVDF layer on a stainless-steel cantilever with neodymium magnets and planar coils. Walking vibrations flexed the cantilever and generated power through the piezoelectric effect. The study demonstrated a stable output voltage in the range of approximately 1–1.5 V with current values around 0.2–0.4 mA and power output between 200–400 μ W. More importantly for this project, it confirmed that a cantilever structure embedded in footwear can deliver stable output even at the slow, uneven pace of normal walking. That is the architectural idea we adopted as our starting point.

Kaushik Pal (2017) reviewed triboelectric nanogenerators as a class of energy harvesters. These devices use materials like PTFE and PET, where charge is generated by contact and separation — essentially friction-based generation. The study reported open-circuit voltages in the range of 60–90 V and short-circuit currents of approximately 1–3 μ A, with transferred charge in the nano-coulomb range. The catch is the high internal impedance, which means the actual current delivered to a load is tiny. Without a well-designed power conditioning circuit, those voltage figures do not translate into usable power.

Mahapatra et al. (2020) took a different material route, embedding ZnO nanoparticles into a PVDF film to boost its piezoelectric response. The device demonstrated voltage outputs in the range of 1–3 V and current values around 0.1–0.3 mA, with power output between 10–100 μ W. The study did note, though, that single-mechanism piezoelectric systems hit a ceiling fairly quickly, and that the energy they produce needs careful management to be practically useful.

Wang et al. studied triboelectric generators using PDMS and PET layers with copper and ITO electrodes. It demonstrated high voltage generation in the range of 100–200 V under foot motion. However, similar to other triboelectric systems, the current output remains in the microampere range due to high impedance. Their conclusion was similar to Pal's review: high voltage generation is achievable but getting that energy into a form a circuit can actually use demands additional conversion and storage circuitry.

Looking across these studies, a pattern emerges. Piezoelectric systems produce moderate voltages with reasonable current, making them more directly usable. Triboelectric systems hit higher voltages but struggle with current delivery. Hybrid systems try to get the best of both worlds, but that comes with more complexity and more components to go wrong.

Given all of this, the decision to go piezoelectric-only for this project makes sense. The LDT2-028K sensor on a cantilever gives a consistent output at walking frequencies. Since the generated output is low (approximately 1–1.2 V and microampere current), additional circuits such as a bridge rectifier and a MOSFET-based switching system are incorporated to efficiently convert, store, and utilize the harvested energy. This approach ensures a simple, low-cost, and practical solution for wearable energy harvesting applications.

2.2 RELATED PRODUCTS

In the field of wearable and vibration-based energy harvesting, several commercial products and research-grade devices have been developed to convert mechanical energy into electrical energy. These systems primarily utilize piezoelectric materials and vibration-based mechanisms, making them relevant to footwear-based energy harvesting applications.

2.2.1. MIDE Vulture™ Piezoelectric Energy Harvester (V25W)

The MIDE Vulture V25W is a widely used piezoelectric energy harvester designed to convert mechanical vibrations into electrical energy. It is commonly used in industrial monitoring and low-power electronic systems.

KeyFeatures:

- Piezoelectric cantilever structure designed for vibration energy harvesting.
- Generates AC voltage when subjected to bending or mechanical stress.
- Compact and lightweight design suitable for integration into wearable systems.
- Operates effectively at low-frequency vibrations similar to human walking motion.

This device demonstrates the effectiveness of cantilever-based piezoelectric harvesting, similar to the approach used in this project.

2.2.2. Advanced Cerametrics PPA Series Piezoelectric Generators

The PPA series from Advanced Cerametrics consists of piezoelectric power generators designed for harvesting energy from dynamic mechanical inputs such as vibration and pressure.

Key Features:

- High-output piezoelectric materials optimized for energy harvesting.
- Suitable for low-power applications such as wireless sensors and IoT devices.
- Provides AC output requiring rectification and energy storage circuits.
- Available in different configurations for structural and wearable applications.

These generators highlight the practical use of piezoelectric materials for energy harvesting in real-world applications.

2.2.3. Power cast P2110B Power harvester Receiver

The Power cast P2110B is a commercial energy harvesting module designed to capture and convert RF energy into usable DC power. Although it primarily uses RF energy, it is often integrated with other energy harvesting systems for powering low-power electronics.

KeyFeatures:

- Converts ambient energy into DC output for electronic loads.
- Includes power management and voltage regulation circuitry.
- Designed for ultra-low-power applications such as sensors and wearable devices.
- Supports energy storage using capacitors or rechargeable batteries.

This product emphasizes the importance of efficient power management, which is a key aspect of the proposed system in this project.

From the above products, it is evident that piezoelectric energy harvesting systems are widely used for low-power applications. However, most existing systems focus either on energy generation or power conditioning separately. Therefore, this project integrates a piezoelectric cantilever, rectifier circuit, and MOSFET-based switching system to efficiently generate, store, and utilize energy from human walking motion in a compact and cost-effective manner.

CHAPTER 3

MATERIALS

3.1 HARDWARE COMPONENTS

3.1.1 Piezoelectric Sensor (LDT2-028K)



Figure 3.1 - LDT2-028K

Purpose: To convert mechanical vibrations generated during walking into electrical energy.

Working: The piezoelectric sensor operates on the piezoelectric effect, where mechanical deformation produces an electrical charge. When the cantilever bends due to walking-induced vibrations, the sensor generates an alternating voltage output.

Key Features & Specifications:

Output Voltage: ~10 mV – 100 V (depends on applied force and circuit conditions)

Observed Output (Practical): ~1 – 1.2 V (open circuit)

Current: Microampere range

Impedance: Minimum 1 M Ω (high impedance device)

Frequency Response: Suitable for low-frequency vibrations (walking motion)

3.1.2 Cantilever Structure

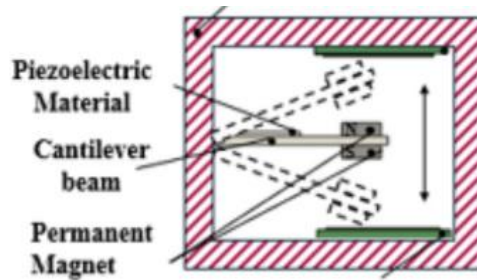


Figure 3.2 – cantilever

Purpose: To amplify mechanical vibrations and enhance the deformation of the piezoelectric sensor for improved energy generation.

Working: The cantilever is fixed at one end and free at the other. When subjected to vibrations from walking, it bends and oscillates, increasing the strain experienced by the piezoelectric sensor mounted on it.

Key Features & Specifications:

Material: Flexible metal/plastic strip

High sensitivity to vibration

Simple and low-cost design

3.1.3 Bridge Rectifier (DB107)



Figure 3.3 – DB107

Purpose: To convert the AC output generated by the piezoelectric sensor into DC voltage.

Working: The DB107 bridge rectifier uses four diodes arranged in a bridge configuration to allow current flow in one direction, converting alternating voltage into pulsating DC.

Key Features & Specifications:

Type: Full-wave bridge rectifier
Output: Pulsating DC
Compact and reliable
Suitable for low-power applications

3.1.4 Capacitor (Energy Storage)

Purpose: To store the rectified DC energy generated by the piezoelectric sensor.

Working: The capacitor stores electrical energy in the form of an electric field and smooths the pulsating DC output from the rectifier.

Key Features & Specifications:

Type: Electrolytic capacitor
Capacitance: 2200 μF
Voltage Rating: 25 V
Function: Energy storage and voltage smoothing

3.1.5 MOSFET Switching Circuit

Purpose: To manage power delivery between the energy harvester and the rechargeable battery.

Working: The MOSFET-based switching circuit monitors the capacitor voltage. When the voltage exceeds 2.4 V, the load is powered using harvested energy. If the voltage falls below this threshold, the system automatically switches to the rechargeable battery.

Key Features & Specifications:

Type: MOSFET-based switching
Threshold Voltage: ~ 2.4 V
Low power loss
Efficient power control

3.1.6 Rechargeable Battery

Purpose:

To provide backup power when the harvested energy is insufficient.

Working:

The battery supplies power to the load when the capacitor voltage is below the required threshold.

Key Features:

Type: Panasonic rechargeable battery

Capacity: **800 Mah**

Provides stable backup power

Ensures continuous system operation

3.2 SOFTWARE COMPONENTS

3.2.1 LTspice

Purpose: LTspice is used for simulating the electrical behaviour of the energy harvesting circuit before hardware implementation.

Working: The designed circuit, including the piezoelectric source, bridge rectifier, capacitor, and load, is modelled in LTspice. The software performs transient analysis to observe voltage and current variations over time. It helps in verifying the rectification process and energy storage performance.

Key Features & Specifications:

- Analog circuit simulation tool developed by Analog Devices
- Supports transient, AC, and DC analysis
- Used to analyse rectifier output and capacitor charging behaviour
- Helps in optimizing component values before hardware implementation

3.2.2 KiCad

Purpose: KiCad is used for designing the Printed Circuit Board (PCB) of the energy harvesting system.

Working: The schematic of the circuit is first created using KiCad's schematic editor. Components such as the rectifier, capacitor, and MOSFET switching circuit are connected virtually. The design is then converted into a PCB layout, where component placement and routing are performed to create a compact and functional board.

Key Features & Specifications:

- Open-source PCB design software
- Supports schematic capture and PCB layout design
- Enables 2D and 3D visualization of PCB
- Provides tools for routing, footprint management, and design verification
- Suitable for creating compact and reliable circuit boards

CHAPTER 4

METHODOLOGY

The system we designed is essentially a chain of four operations: capture the energy from walking, clean up the signal, store it, and then decide how to use it. A piezoelectric sensor on a cantilever handles the first part, a bridge rectifier deals with the signal, a capacitor stores the charge, and a MOSFET switching circuit manages delivery to the load.

When someone walks, each footstep sends a vibration through the shoe. That vibration travels into the cantilever and makes it flex. The LDT2-028K sensor, being bonded to the cantilever, gets stretched and compressed as the beam bends, and that mechanical strain is what generates the alternating voltage through the piezoelectric effect.

The AC output from the sensor goes straight into a DB107 bridge rectifier, which flips the negative half-cycles into positive ones — turning the alternating signal into a pulsating DC. From there, the charge accumulates in a 2200 μF , 25 V electrolytic capacitor. The capacitor both smooths the ripple and holds the harvested energy between footsteps.

Once there is enough stored charge, the MOSFET switching circuit takes over. It keeps watching the capacitor voltage, and as soon as that voltage crosses 2.4 V, it connects the load to the harvester side. If the voltage drops below that — say, during a rest or a slow walk — it switches the load over to the rechargeable battery automatically, so there is no interruption.

Everything — the rectifier, capacitors, MOSFETs, resistors, and connectors — is laid out on a single PCB designed in KiCad. Putting it all on one board keeps the form factor small and the connections reliable, which is important for anything meant to go into or onto a shoe.

4.1 WORKING

The working of the system can be explained in the following steps:

1. Energy Generation:

Foot impacts during normal walking send low-frequency vibrations into the shoe. These vibrations flex the cantilever, and the sensor bonded to it produces an alternating voltage.

2. Rectification:

That AC output goes through the DB107 bridge rectifier, which converts it to pulsating DC.

3. Energy Storage:

The rectified signal charges up the 2200 μF storage capacitor, which also smooths out the ripple in the process.

4. Power Management (Switching):

A MOSFET-based switching circuit monitors the capacitor voltage:

- If voltage $> 2.4\text{ V} \rightarrow$ Load is powered by harvested energy
- If voltage $< 2.4\text{ V} \rightarrow$ Load is powered by rechargeable battery

5. Load Supply:

Eventually, the regulated output powers the load — low-power devices like step counters or motion sensors, which will be integrated in the next version of the system.

4.2 BLOCK DIAGRAM

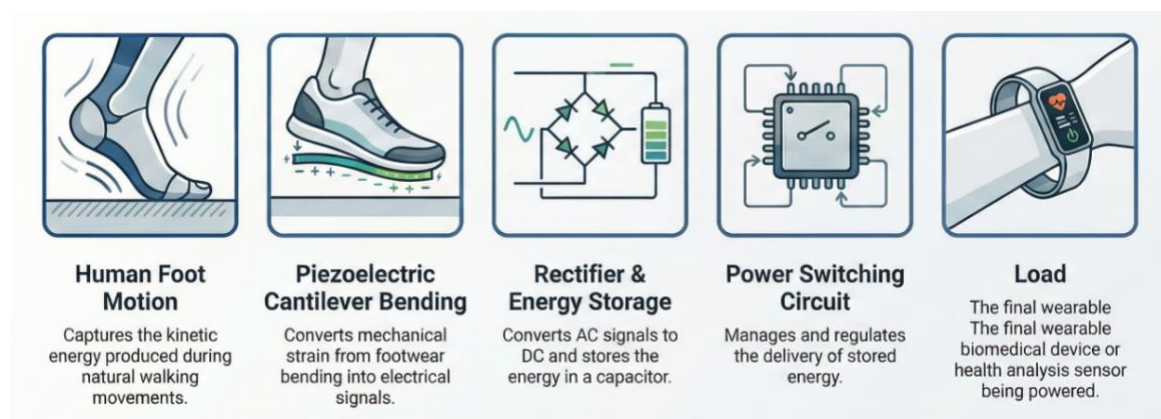


figure 4.1 block diagram

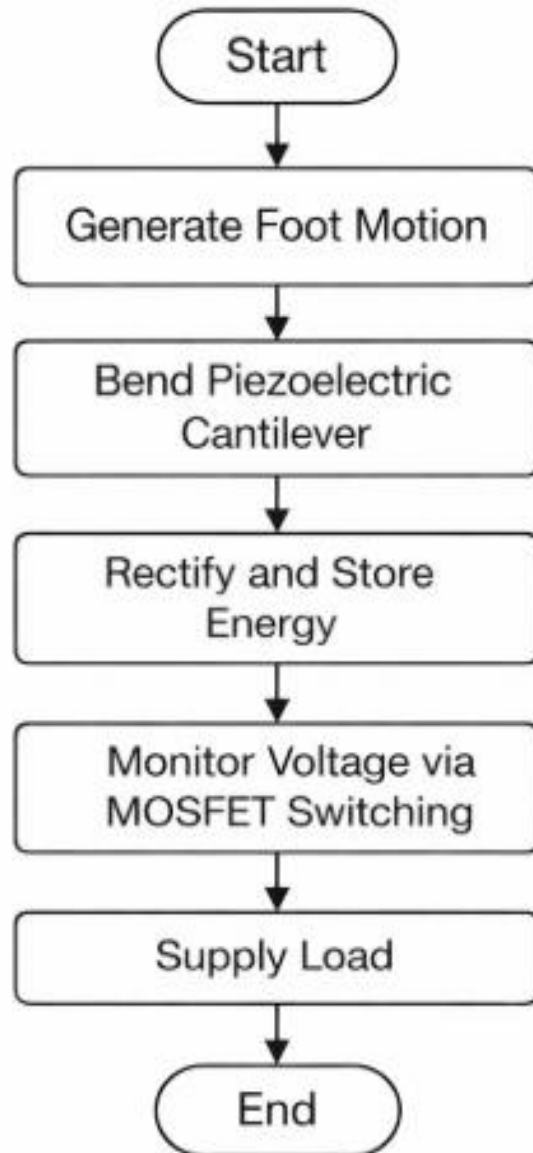


figure 4.2 flow chart

1. Energy Generation Stage:

During this stage, mechanical energy is generated from human walking motion. When a person walks, the foot produces low-frequency vibrations. These vibrations act as the primary input to the system and serve as the source of energy for the harvesting process.

2. Piezoelectric Conversion Stage:

The generated mechanical vibrations cause the cantilever structure to bend. The

piezoelectric sensor (LDT2-028K) mounted on the cantilever experiences mechanical strain due to this bending. As a result, the piezoelectric material converts the mechanical energy into an alternating electrical voltage based on the piezoelectric effect.

3. Rectification and Energy Storage Stage:

The AC voltage generated by the piezoelectric sensor is passed through a bridge rectifier (DB107), which converts it into pulsating DC. This DC output is then stored in a capacitor (2200 μ F, 25 V). The capacitor also smooths the voltage, ensuring a more stable DC output for further use.

4. Power Management (Switching) Stage:

In this stage, a MOSFET-based switching circuit monitors the voltage across the capacitor. If the capacitor voltage exceeds the threshold value of 2.4 V, the load is powered using the harvested energy. If the voltage falls below this threshold, the system automatically switches to a rechargeable battery (800 Mah), ensuring uninterrupted operation.

5. Load Supply Stage:

Finally, the stored energy is supplied to the load. The load can include low-power devices such as step counters or motion sensors (to be implemented in future work). This stage represents the effective utilization of the harvested energy generated from human walking. The stored electrical energy from the capacitor or rechargeable battery powers the load, demonstrating the practical application of piezoelectric energy harvesting for small electronic devices and wearable systems.

4.2 POWER MANAGEMENT AND SWITCHING LOGIC

One thing that became obvious early on is that piezoelectric output from walking is neither constant nor particularly large. The energy arrives in short bursts whenever the foot lands, and the amount varies with walking speed and surface. Trying to run a load directly from this kind of source would be unreliable. A proper power management stage is what makes the system actually functional.

The approach taken here uses a 2200 μ F capacitor as a buffer — it charges gradually as the person walks and provides a more stable voltage output than the raw rectified signal would. The MOSFET switching circuit then decides, based on that voltage, where the load should be drawing power from.

The switching logic is straightforward. The circuit checks the capacitor voltage and uses 2.4 V as the decision point:

- **Condition 1: Capacitor Voltage > 2.4 V**

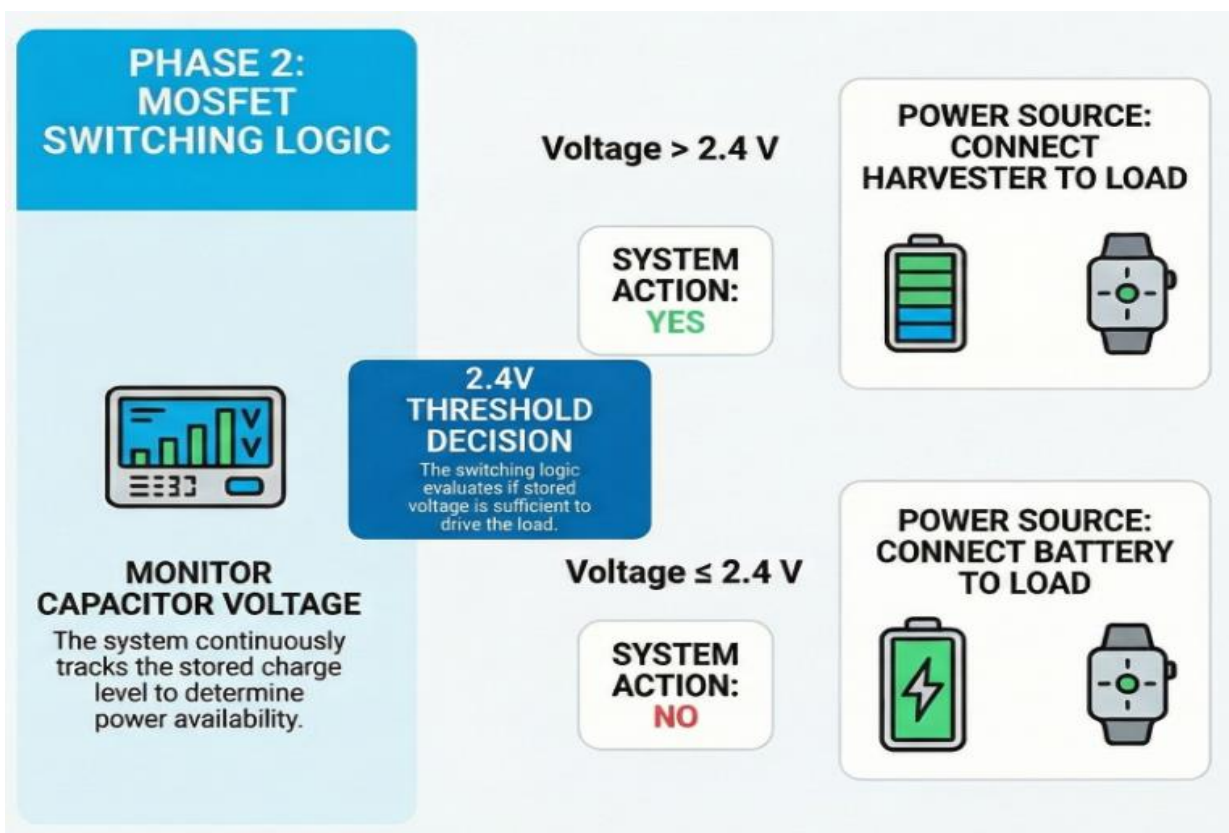
If the voltage is above 2.4 V, the switching circuit connects the capacitor to the load, so the harvested energy is being put to use directly.

- **Condition 2: Capacitor Voltage < 2.4 V**

If it drops below 2.4 V, the circuit disconnects the harvester and brings in the 800 Mah rechargeable battery instead. The load sees a continuous supply either way.

MOSFETs are a good choice for this kind of switching — they have very low on-resistance, so they waste very little energy when conducting, and they switch fast enough that there is no noticeable glitch in the output during transitions.

The end result is a system that makes sensible use of whatever energy is available, falls back gracefully when there is not enough, and never leaves the load without power. For a wearable application where the energy source is as unpredictable as someone's walking pace, that kind of reliability matters.



CHAPTER 5

SYSTEM ARCHITECTURE AND PROTOTYPE DEVELOPMENT

5.1 SYSTEM ARCHITECTURE

Breaking the system down by function, there are four stages that work in sequence. First, mechanical energy from footsteps gets converted into an electrical signal. Then that signal is conditioned — rectified and smoothed — into something stable. After that, the energy is stored until there is enough to be useful. Finally, a power management circuit decides how and when to deliver that stored energy to the load. Each stage feeds into the next, and the overall result is a system that takes something as irregular as a footstep and turns it into a reasonably steady supply for low-power electronics.

The energy input happens at the cantilever, where the LDT2-028K sensor is bonded to a flexible strip mounted on the shoe. When the foot hits the ground, the cantilever deflects and the piezoelectric film generates an AC voltage. That signal is too small and too inconsistent to directly drive anything, so it moves into the rectification stage. It is low in magnitude—typically around 1 to 1.2 V open circuit—and cannot directly power any useful load. So, it first goes through a rectification stage.

Rectification is handled by the DB107, a four-diode full-wave bridge that converts the alternating output into a pulsating DC signal. The main storage capacitor is 2200 μF rated at 25 V — large enough to accumulate charge across several footsteps and provide a useful amount of energy at a time. A smaller 10 μF capacitor sits in parallel to catch high-frequency noise that the bulk cap would miss.

The power management block uses a voltage divider — two 470 k Ω resistors — to sample the capacitor voltage and apply a scaled version to the gates of two IRLML6402 P-channel MOSFETs (Q1 and Q2). These two transistors handle the switching between the harvested energy path and the backup battery. When the harvested voltage is sufficient, the MOSFETs route power from the capacitor to the load through output connector J3. When it is not, they switch in the battery. The output rail at J3 stays alive throughout.

The schematic captures how all four blocks are wired together — it is worth looking at in detail to understand the signal and power paths.

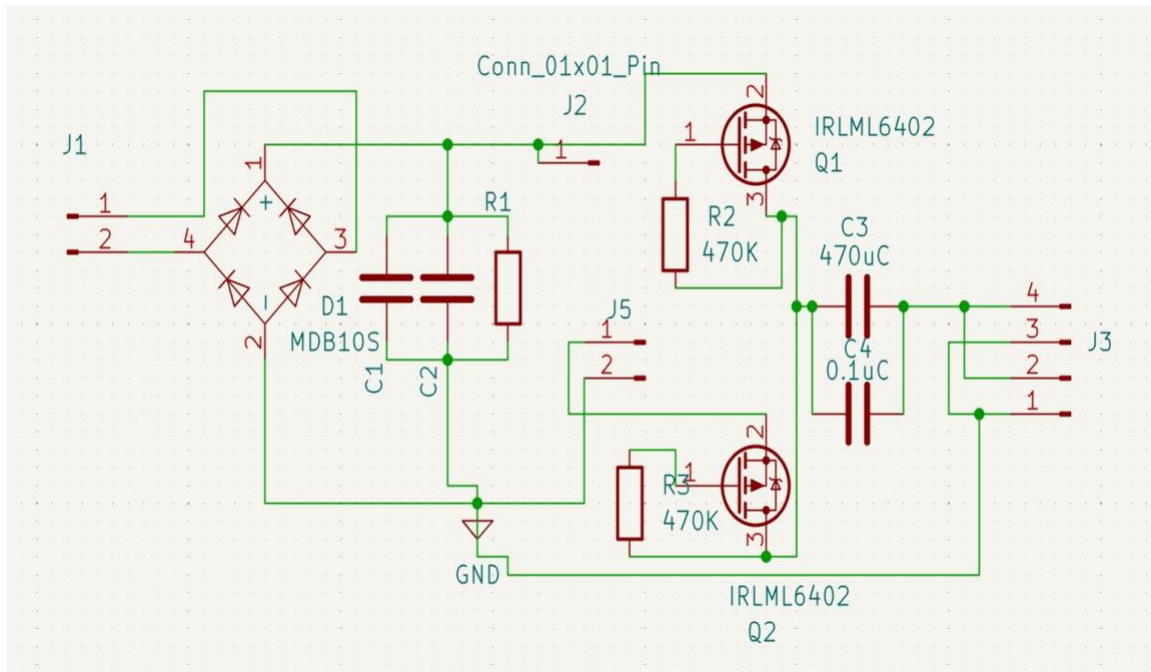


Figure 5.1 – Circuit Schematic of the Energy Harvesting System

5.2 PROTOTYPE DEVELOPMENT

We did not jump straight to a PCB. The development went through stages — starting on a breadboard, testing each section individually, then putting things together before finally committing the design to a board. That approach caught a few problems early that would have been annoying to debug on a finished PCB.

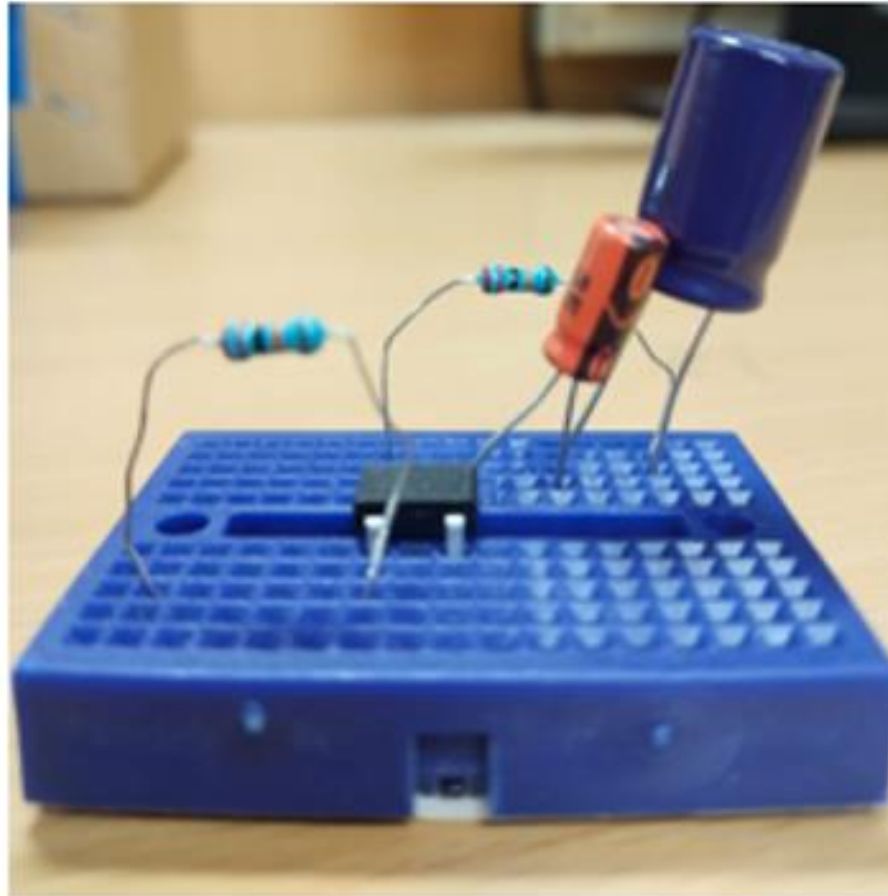
The first thing we tested was just the sensor on the cantilever, with nothing else connected. We simulated walking vibrations by bending the strip repeatedly and measured the open-circuit voltage using a multimeter and an oscilloscope. The output peaked in the 1 to 1.2 V range, which was consistent with what the literature suggested for this type of sensor. That confirmed the mechanical setup was working.

Next, we added the DB107 rectifier to the breadboard and connected it to the sensor. Several capacitor values were tried in the storage position to see how well they accumulated charge. The storage capacitor (2200 μ F capacitor was clearly the best option — it built up voltage steadily across several bending cycles and held it long enough to be useful. We monitored the voltage over time to get a sense of the charging curve.

The switching circuit came next. We wired up the two IRLML6402 MOSFETs with 470 k Ω resistor dividers at the gates and used a bench power supply to manually sweep the input voltage up and down through the 2.4 V threshold. A resistive dummy load let us

observe the output during transitions. The switching happened cleanly at the expected point, with no obvious glitching.

Once each section checked out individually, we connected everything together and ran a real walking test. The sensor was placed in the shoe, the circuit was attached, and we walked around for a while monitoring the output. The system behaved the way it was supposed to — energy accumulated in the capacitor during walking, the switching triggered when the voltage crossed 2.4 V, and the load stayed powered throughout. That was the point where we felt confident enough to move on to the PCB.



[Figure 5.2 – Rectifier Prototype assembled on breadboard]

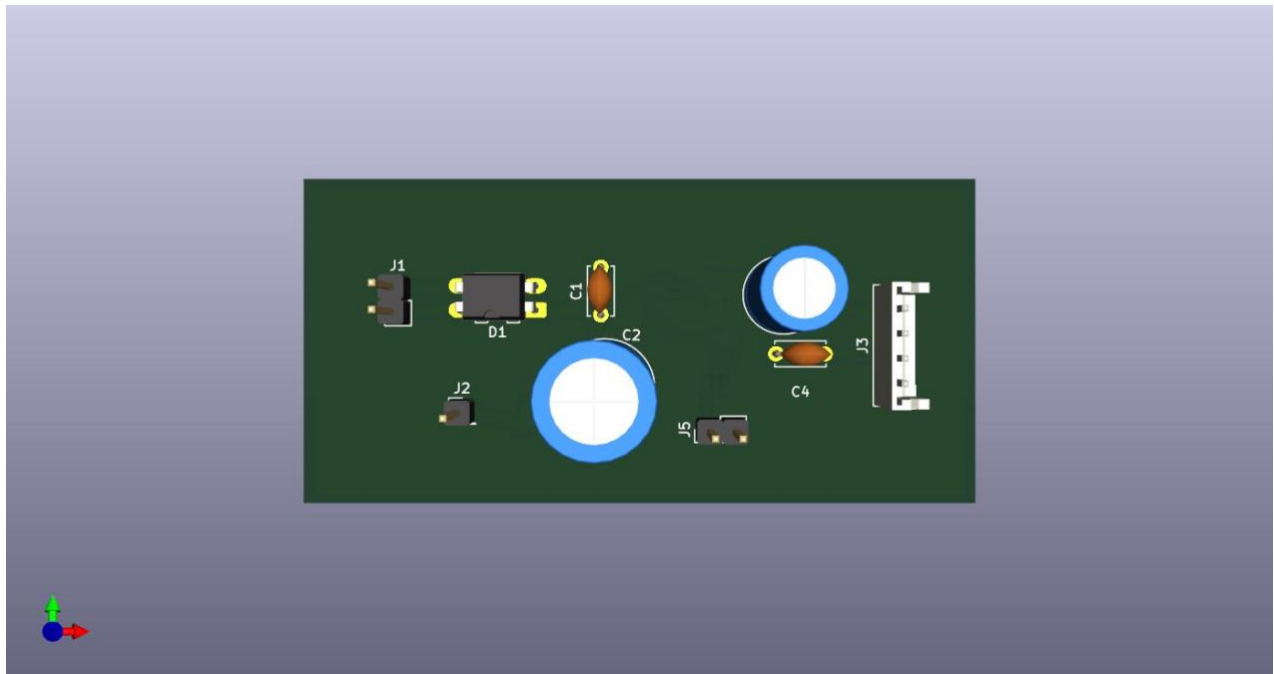


Figure 5.3.1 – Fabricated PCB (KiCad View)

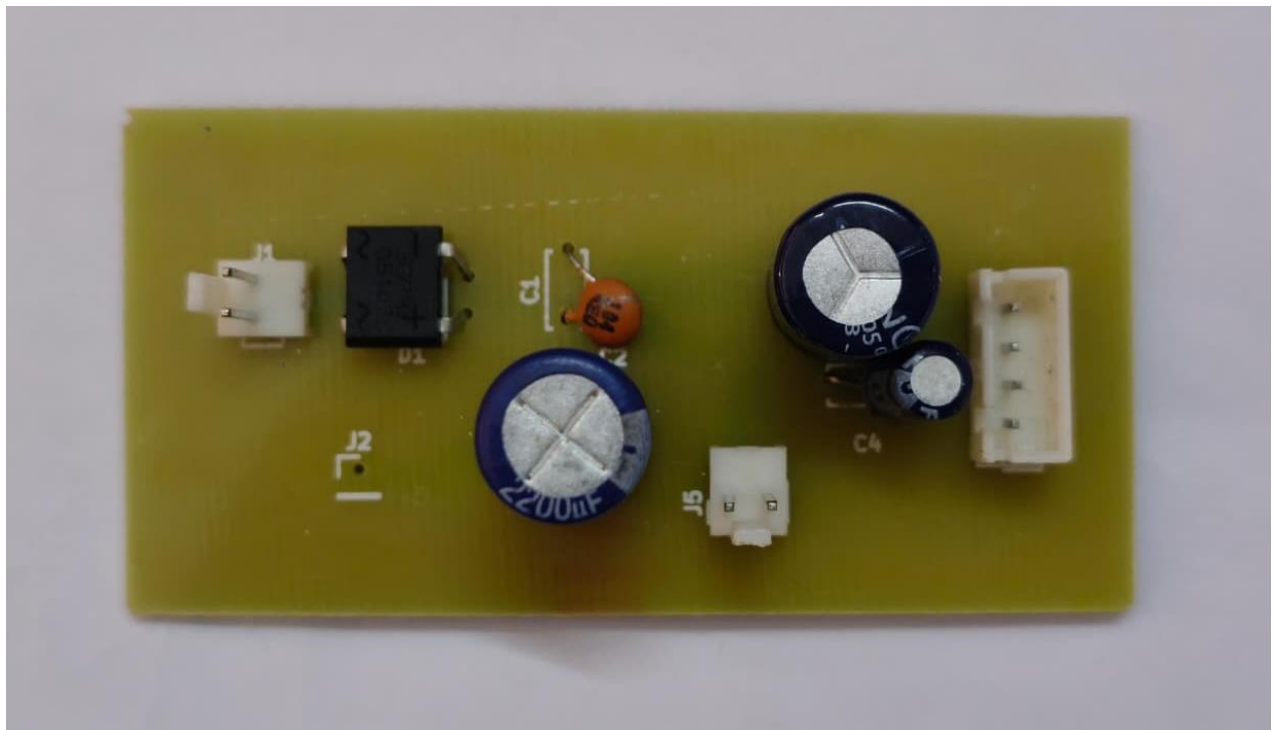


Figure 5.3.2 – Fabricated PCB (Prototype)

5.3 PCB DESIGN AND IMPLEMENTATION

With the breadboard prototype working, the next step was designing a proper PCB. The goal was to get everything onto a single board that was compact enough to be worn comfortably, reliable enough to survive real-world use, and laid out in a way that would not introduce new problems through poor routing or component placement.

The schematic was first drawn in KiCad's schema tool, making sure all the nets were named and connected correctly. When it came to footprints, we used a mix of SMD and through-hole. The MOSFETs (SOT-23) and resistors (1206 metric) are SMD, which keeps them small. The capacitors, DB107 rectifier, and connectors are through-hole, which makes them easier to hand-solder and swap out during testing. That combination worked well for a one-off prototype.

The PCB layout was then designed in KiCad's Pcb new environment. Component placement was guided by signal flow—the rectifier and filter capacitors were placed on the left side near the piezoelectric input connector (J1), while the MOSFET switching section was positioned towards the center, and the output connector (J3) was placed on the right. That layout keeps power traces short and makes the board easier to troubleshoot. Ground planes were poured on both copper layers to keep the board well-grounded and reduce any noise that might interfere with the low-level signals coming from the piezoelectric sensor.

Trace widths were sized for the currents in each section. The main power path from the rectifier to the storage capacitor uses 1 mm traces, while the gate resistor network, which carries only a tiny signal current, uses 0.25 mm traces. The board outline is a simple rectangle — nothing fancy — but the dimensions were chosen with an eventual enclosure in mind.

The BOM is short — DB107 rectifier, two IRLML6402 MOSFETs, a handful of capacitors, resistors, and pin headers. None of these parts are expensive, which was an explicit goal. The board was assembled by hand-soldering, then checked for shorts and open connections with a multimeter before being connected to the sensor for functional testing.

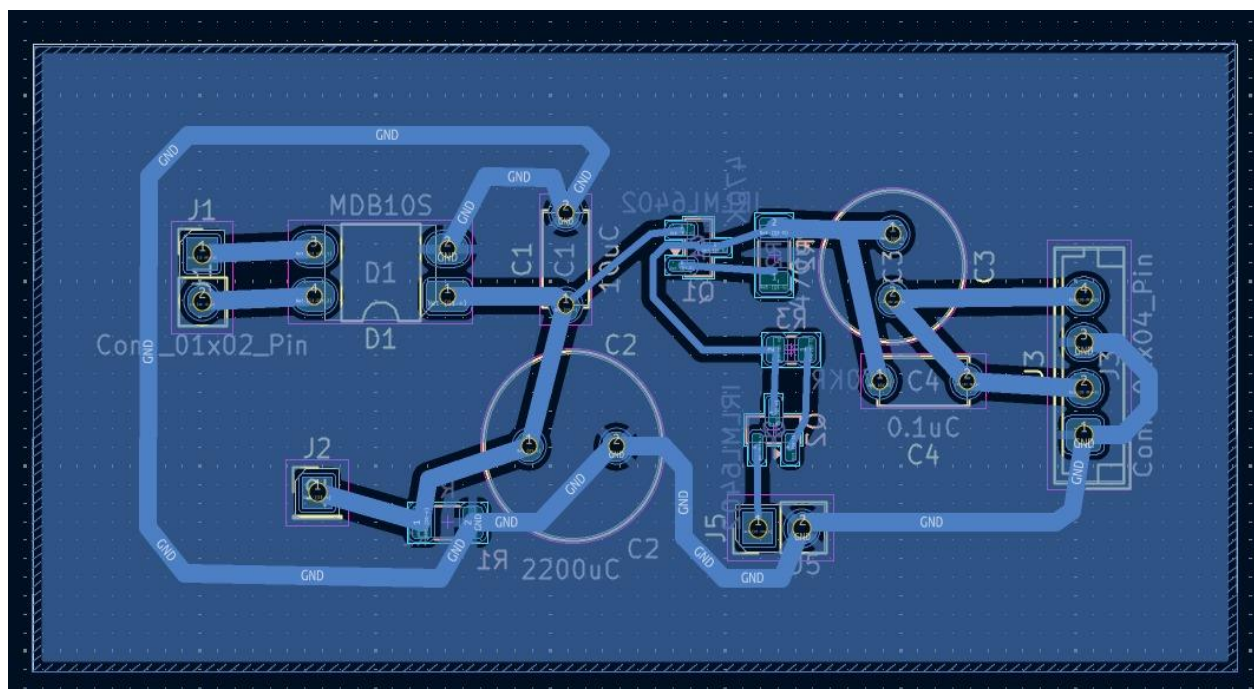


Figure 5.4.1 – PCB Layout (KiCad 2D View)

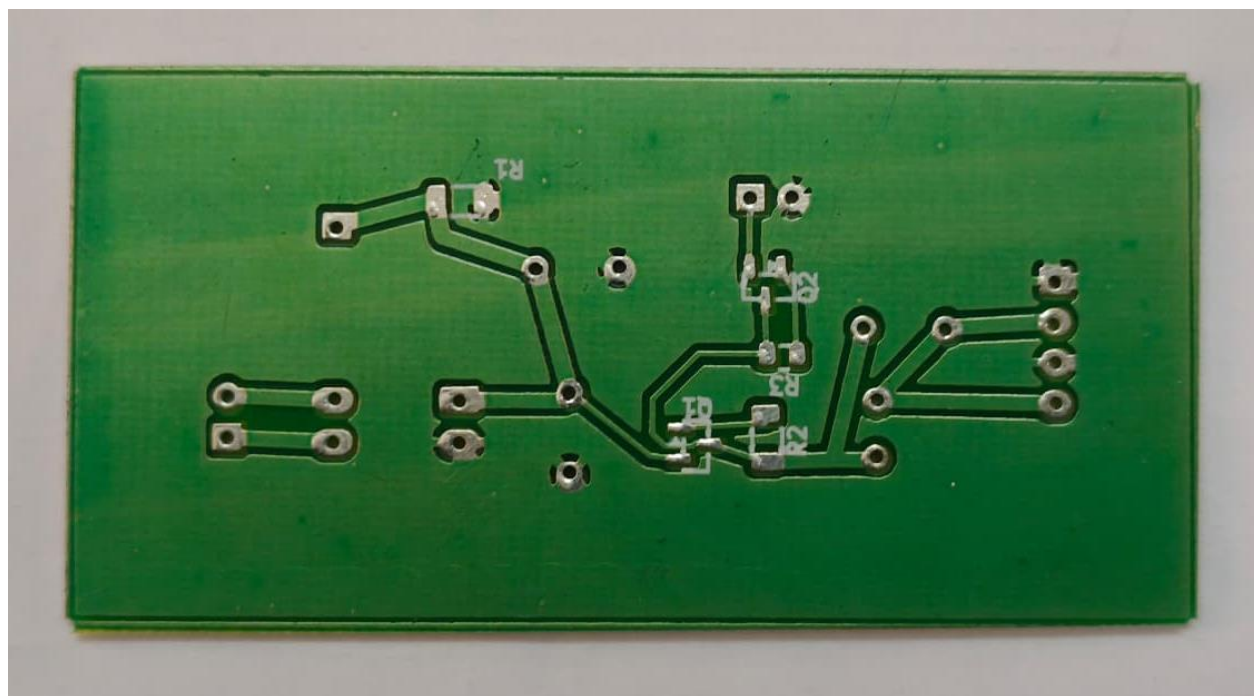


Figure 5.4.2 – PCB Layout (Prototype)

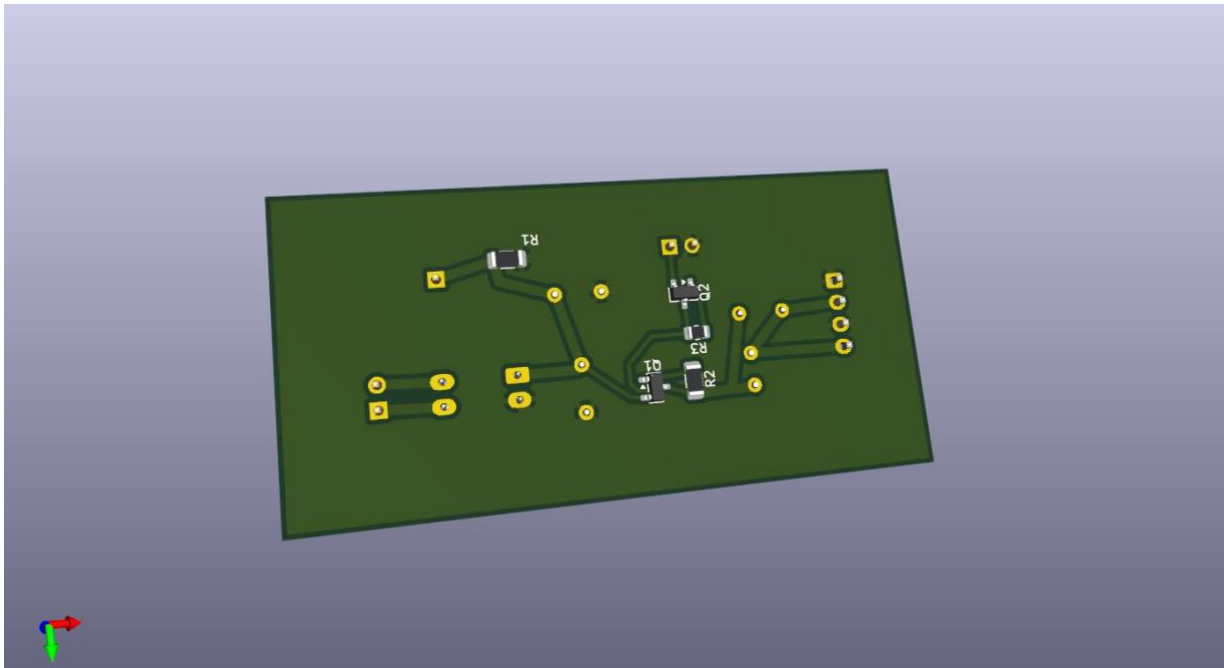


Figure 5.5.1 – PCB 3D Render (KiCad)

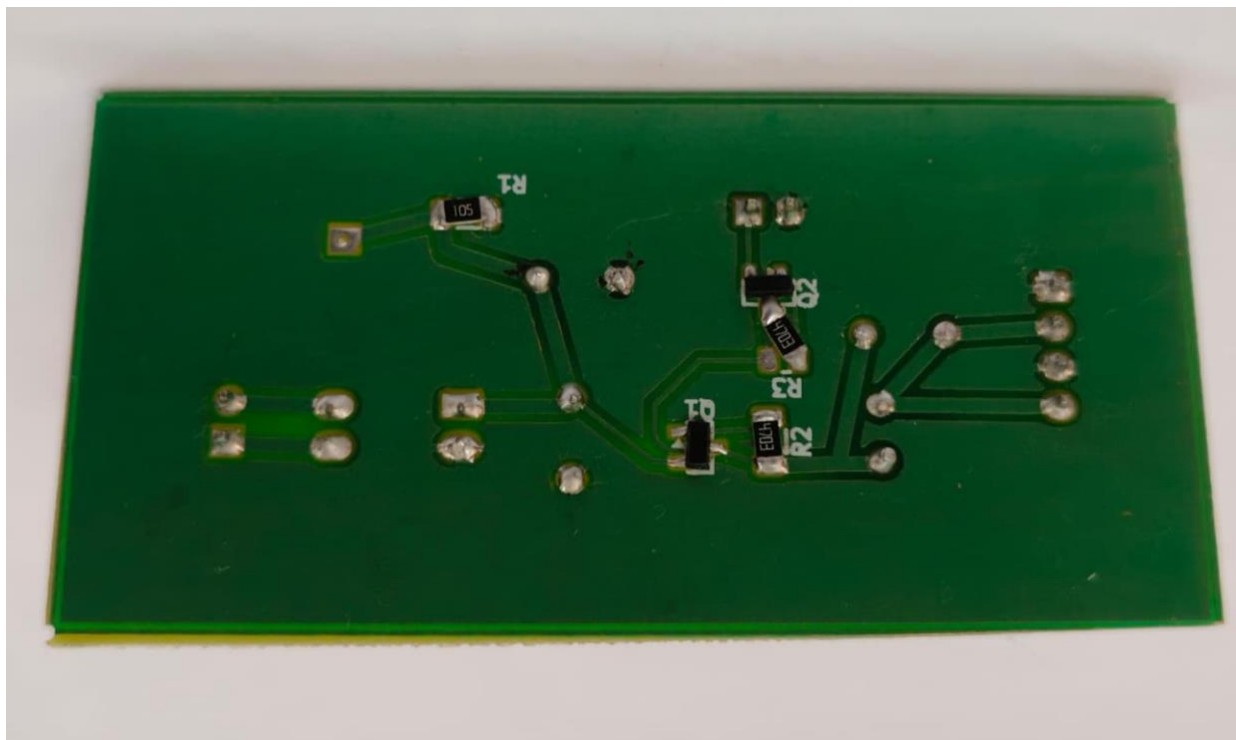


Figure 5.5.2 – PCB (prototype)

CHAPTER 6

RESULTS AND DISCUSSION

6.1 EXPERIMENTAL RESULTS

Testing the complete system involved walking under consistent conditions and measuring what was happening at each stage using a multimeter and oscilloscope. The goal was to see whether the actual behavior matched what the design predicted — and to identify anything that did not behave as expected.

Piezoelectric Sensor Output

The sensor output varied noticeably depending on how hard the foot landed and how fast we were walking. Open-circuit voltage ranged from about 0.7 V at a gentle pace up to 1.2 V with harder footfalls. The waveform on the oscilloscope looked irregular — not a clean sine wave, which is what you would expect from something driven by human gait rather than a machine. Current stayed in the microampere range throughout, which is typical for a high-impedance PVDF sensor.

Rectifier Output

After rectification, the output measured between about 0.5 V and 0.9 V — lower than the raw sensor output because of the forward voltage drop across the diodes. That drop was expected and confirms the rectifier is working correctly. The pulsating shape of the DC signal also showed up clearly on the oscilloscope, which is consistent with proper full-wave rectification.

Capacitor Charging Behavior

The 2200 μF storage capacitor was observed to accumulate charge gradually with each walking cycle. Under continuous walking at a moderate pace (approximately 80–100 steps per minute), the capacitor voltage rose steadily over time. In a typical test session, the voltage across the capacitor reached approximately 1.8 V to 2.1 V after about 60 to 90 seconds of walking. Reaching the 2.4 V switching threshold required a longer duration or higher intensity of walking, which reflects the inherent low-power nature of the piezoelectric source.

Switching Circuit Behavior

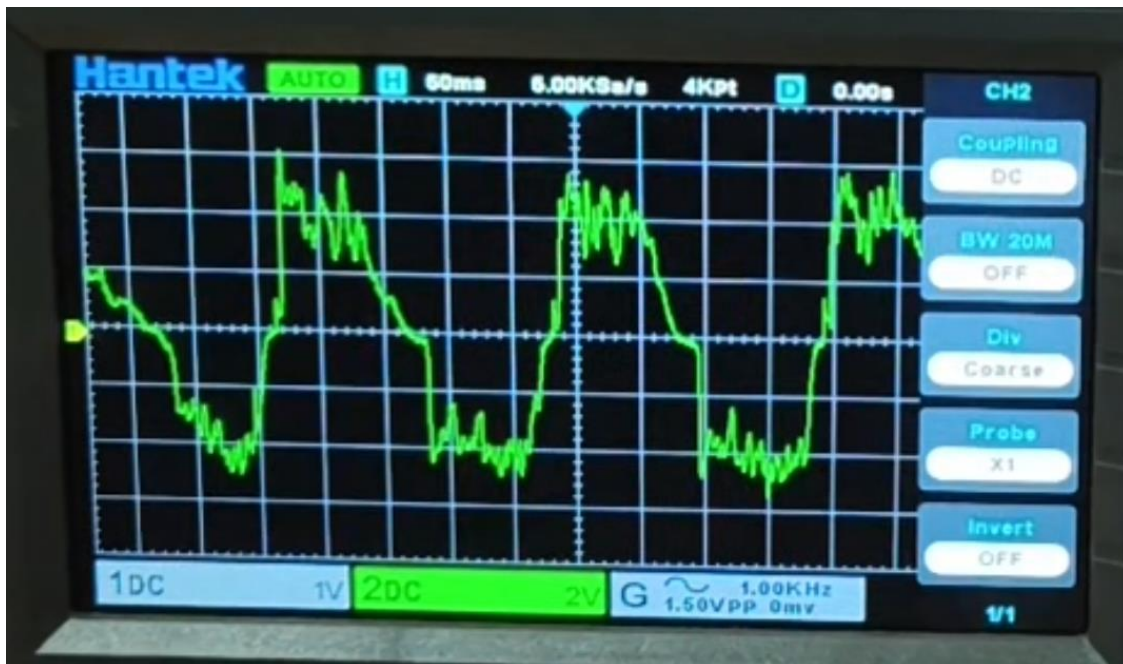
We also tested the switching circuit in isolation using a bench power supply to control the voltage at the capacitor terminals. Sweeping through the 2.4 V threshold showed Q1 and Q2 switching cleanly in both directions. There was no noticeable drop or spike at the output during the transition — the voltage at J3 stayed stable. There was a small amount

of ripple visible when the switching happened, but nothing that would cause a practical problem for the kind of low power loads this system is meant to drive.

Hybrid Output Measurement

Pulling together the measurements from all stages, here is what the system produced:

- Piezoelectric open-circuit voltage: 0.7 V – 1.2 V (AC)
- Rectified DC output: ~0.5 V – 0.9 V (pulsating)
- Capacitor voltage (after ~90 seconds of walking): ~1.8 V – 2.1 V
- Estimated harvested power: ~170 – 220 μW
- MOSFET switching threshold: 2.4 V (verified)
- Load output voltage: Stable during both harvester-powered and battery-powered modes



[Figure 6.1 – Oscilloscope capture: Piezoelectric AC output waveform]

[Figure 6.2 – Capacitor charging curve over walking duration]

6.2 DISCUSSION

The results hold up reasonably well against what was expected going in. The measured power output of 170 to 220 μW is in the same ballpark as similar cantilever-based piezoelectric systems reported in the literature — the work by Iftikhar Ahmad et al. (2025) used a comparable PVDF cantilever approach and got similar figures, which gives us some confidence that the measurements are in the right range.

One thing that stood out during testing was how sensitive the output is to be walking style and sensor placement. Faster walking and harder footfalls produced meaningfully higher voltages than a gentle stroll. The implication is that the cantilever geometry and its position in the shoe are worth optimizing — there is probably more energy available than the current setup is capturing.

The diode drop in the bridge rectifier is worth flagging as a limitation. Losing 0.3 to 0.7 V per diode across a two-diode path adds up when the source voltage is already below 1.2 V. A synchronous rectifier or Schottky diodes with lower forward voltages would help recover some of that lost margin. It is not a critical problem for the prototype, but it is the first thing we would address in a refined design.

The capacitor charged more slowly than we initially estimated, mostly because the piezoelectric source is high impedance and only delivers energy during the brief instant of bending. 2200 μF is quite a large capacitor to charge from such a limited source, so reaching 2.4 V takes longer than you might expect. In real use, there would be a warm-up period at the start of a walk before the system switches into harvester-powered mode. That is not really a problem since the battery backup covers that gap, but it is worth being transparent about.

The switching circuit was the part of the system we were most unsure about before testing, and it turned out to be the most reliable. Transitions between the harvester and battery were smooth, the output stayed stable, and the IRLML6402 MOSFETs handled the job without any sign of stress. Their low on-resistance keeps losses minimal, and the SOT-23 package fits neatly on the PCB.

Taking everything together, the system works. It is not going to power anything demanding, and 170 to 220 μW is an honest description of what piezoelectric shoe harvesting realistically delivers. But for the kinds of loads it is intended for — step counters, low-power sensors, occasional short transmissions — that is genuinely enough. The prototype gives a solid base to build from, and the next stage of work, connecting real loads and refining the circuit, has a clear path forward.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

Looking back on the work, the core goal was achieved. A complete energy harvesting system — from cantilever sensor to power management circuit — was designed, built, and tested. The LDT2-028K piezoelectric sensor, DB107 rectifier, 2200 μF storage capacitor, and IRLML6402 MOSFET switching circuit are all working together on a single KiCad-designed PCB.

The measured results were honest: 0.7 to 1.2 V open circuit from the sensor, a rectified output in the 0.5 to 0.9 V range, and an estimated delivered power of 170 to 220 μW after storage and switching. The 2.4 V threshold switching between harvested energy and battery backup worked reliably, and the load remained powered throughout all tests.

What this project demonstrates, more than anything, is that self-powered wearable sensing from footstep energy is not just theoretically possible — it is practically achievable with inexpensive, off-the-shelf parts and open-source tools. The energy levels are modest, but they are real, and for the right kind of load, they are enough. That positions this work as a practical steppingstone toward wearable biomedical systems that do not depend on frequent battery maintenance.

7.2 FUTURE WORK

The prototype shows that the concept works, but there is plenty of room to push it further. Several specific directions are worth pursuing in future work.

Load Integration: The first priority is connecting actual loads — a step counter and an MPU sensor are the obvious candidates. Both draw very little current when in sleep or low-power mode, which suits the intermittent nature of harvested energy well. Testing with real loads will give a much clearer picture of how the system performs in a genuine use scenario.

Improved Rectification: The diode drop in the current bridge rectifier is a meaningful loss given how low the source voltage is. Schottky diodes or a synchronous rectifier would reduce that loss significantly and make more of the harvested energy available for storage. This is a straightforward upgrade with a clear payoff.

Dedicated Energy Harvesting IC: The discrete MOSFET switching circuit works, but it is fairly basic. ICs like the LTC3588 or BQ25504 are designed specifically for piezoelectric sources and include maximum power point tracking and regulated output in a single package. Switching to one of those would improve efficiency and simplify the design considerably.

Cantilever Optimization: The cantilever dimensions were chosen based on available materials and intuition rather than rigorous analysis. Running FEA simulations to tune the length, width, and thickness for the 1 to 2 Hz walking frequency band should improve energy capture meaningfully.

Hybrid Energy Harvesting: Adding a triboelectric layer — a PET-PTFE TENG built into the insole — alongside the existing piezoelectric cantilever could capture energy from the sliding and contact motion that the cantilever misses. A hybrid system would likely produce enough combined output for continuous sensor operation rather than just intermittent bursts.

Miniaturization and Enclosure: The PCB needs a proper housing before it can be considered wearable in any practical sense. A molded enclosure that keeps out sweat and handles the mechanical stress of being in a shoe is the next engineering challenge. Keeping that enclosure lightweight will also matter for user comfort.

Wireless Data Transmission: The longer-term goal is a system that transmits health data — step count, cadence, gait patterns — wirelessly over BLE or a similar low-power protocol, all from harvested energy alone. That is not achievable with the current power levels, but it is a realistic target once the harvesting efficiency and storage are improved. Getting there would mean a truly battery-free wearable health monitor.

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